Introduction

Motivation

Portfolio construction has been a central problem in quantitative finance for over 70 years, originating with Markowitz (1952)'s mean-variance framework. Despite this long history, published comparison studies typically pit 3–5 methods against each other on a single dataset, making it difficult to draw general conclusions about which approaches survive out-of-sample across different market regimes. The rise of covariance estimator improvements (Ledoit and Wolf 2004; Chen et al. 2010), diversification-based objectives (Choueifaty and Coignard 2008; Maillard et al. 2010), hierarchical risk parity (López de Prado 2016), Black-Litterman views (Black and Litterman 1992; He and Litterman 1999), and volatility-managed overlays (Moreira and Muir 2017) has expanded the strategy universe substantially, yet comprehensive head-to-head evaluations on a common universe and harness remain scarce.

The benchmark proposed by DeMiguel et al. (2009) — showing that naive equal-weight (1/N) outperforms 14 mean-variance models on most standard datasets — raised the bar for demonstrating out-of-sample improvements, but tested only a narrow slice of the available strategy space. Most strategies introduced in the decade since have been evaluated in isolation rather than side-by-side. The need is acute: practitioners face decisions not just about which strategy family to use, but whether shrinkage or sample covariances are better, whether VMP overlays add value uniformly or only for specific base strategies, and whether regime-conditional switching is substitutable with volatility management. These are empirical questions that require simultaneous evaluation on the same data over the same period.

This study expands the comparison to 58 strategies, standardizes the evaluation harness, and systematically adds a VMP overlay to every base method — providing the most comprehensive single-universe allocation comparison we are aware of.

Related Work

The foundational challenge of portfolio out-of-sample performance was crystallized by DeMiguel et al. (2009), who showed that naive equal-weight (1/N) outperforms 14 mean-variance models on most standard datasets, attributing the failure to estimation error in expected returns. Michaud (1989) earlier identified the "error maximization" property of the sample Markowitz tangency portfolio — the unconstrained optimizer amplifies estimation errors and concentrates heavily in assets with erroneously large sample Sharpe ratios. Regularization via linear shrinkage (Ledoit and Wolf 2004) and Oracle Approximating Shrinkage (Chen et al. 2010) address the eigenvalue instability in sample covariance matrices.

Diversification-based strategies that side-step expected return estimation include the Maximum Diversification Portfolio (Choueifaty and Coignard 2008) and Equal Risk Contribution (Risk Parity) (Maillard et al. 2010). Hierarchical Risk Parity (López de Prado 2016) uses machine learning clustering to construct a diversified portfolio without matrix inversion, avoiding the instability of classical mean-variance optimizers. The momentum anomaly, documented by Jegadeesh and Titman (1993) for cross-sectional equities and extended to time-series settings by Moskowitz et al. (2012), provides an alternative signal-based weighting scheme. The low-volatility anomaly (Frazzini and Pedersen 2014) motivates factor strategies based on realized volatility ranking. The Black-Litterman model (Black and Litterman 1992; He and Litterman 1999) combines equilibrium market priors with investor views via Bayesian updating, offering a principled framework for incorporating signals without extreme concentration. Fama-French factor portfolios (Fama and French 1993) extended to the momentum factor provide benchmarks for systematic factor-based allocation.

Volatility-managed portfolios (Moreira and Muir 2017) scale exposure inversely to realized variance, generating improved Sharpe ratios across a wide range of base strategies in equity and bond markets. The present study evaluates all of these approaches simultaneously on a single harness, covering an 18-year window that includes the Global Financial Crisis, the COVID-19 crash, and the 2022 rate-shock bear market.

Contribution

This study makes three contributions to the comparative portfolio evaluation literature:

1. **Scale.** To our knowledge, this is the largest single-universe, single-period comparison of diversified allocation strategies, evaluating 58 distinct methods (24 base strategies, 4 constrained MV variants, and 6 long-short variants, each paired with and without a VMP overlay) against a common benchmark on identical data.
2. **Systematic VMP overlay.** Every base strategy is evaluated both with and without the VMP overlay (Moreira and Muir 2017), enabling a clean measurement of the overlay's marginal contribution across all families and isolating the interaction between structural portfolio construction and dynamic volatility management.
3. **Regime-conditional analysis and cost sensitivity.** A custom v2a regime-switching rule constructed from 8-regime FRED-based macro classification is compared against the full strategy universe. All strategies are evaluated net of 10 bps round-trip transaction costs to assess implementability under realistic market frictions.

Data and Methodology

Universe

The evaluation universe comprises 30 tickers spanning six asset classes: 8 large-cap US equity single stocks, 6 US sector ETFs, 2 broad equity index ETFs, 3 international equity ETFs, 5 fixed-income ETFs (investment-grade and high-yield), and 6 commodity, FOREX, and cryptocurrency instruments. Daily close-of-business prices are sourced from EODHD and cached locally; the sample runs from 2008-01-01 to 2026-04-30 (18.3 years, approximately 4,600 NYSE trading days).

The 2008 start date is deliberate: it encompasses three distinct stress regimes — the Global Financial Crisis (2008–09 to 2009–03), the COVID-19 crash (2020–02 to 2020–04), and the 2022 rate-shock bear market — along with multiple expansion phases. This regime heterogeneity provides a robust out-of-sample testing environment for both risk-based and return-based strategies. The benchmark throughout is the equal-weight (EW, 1/N) portfolio (DeMiguel et al. 2009), rebalanced monthly. All Sharpe ratios are computed as

$$S = \frac{\bar{r} - r_f}{\sigma_r} \times \sqrt{252}$$

with $r_f = 0$ throughout (annualized excess-return Sharpe).

Walk-Forward Harness

All 58 strategies are evaluated through a common walk-forward harness (`aiam.harness.run_horse_race`) with the following fixed parameters:

- **Rebalancing:** monthly, on NYSE business days
- **Prices:** close-of-day; weights summing to 1.0, long-only
- **Covariance lookback:** 252 trading days for all mean-variance-family strategies
- **Risk-free rate:** $r_f = 0$ throughout
- **Benchmark:** Equal-weight (EW) portfolio

No transaction costs are applied in the base harness; a 10 bps sensitivity analysis is performed in Section 3.3. The VMP overlay is applied daily (Section 2.4) and is assumed costless in the base analysis; the implications of this assumption are discussed in Section 3.3.

Backtesting Hygiene

Six practices govern the harness implementation. (1) **Look-ahead prevention** — weights at date t use only data observable up to close(t); they apply to realized returns at $t + 1$. (2) **Feature leakage** — all signals (momentum, vol, regime) are constructed from lagged price history; no asset's return at t enters its own feature at t. (3) **Signal/return alignment** — `held_weights = target_weights.shift(1)` is enforced at the harness level; unlagged application manufactures spurious performance. (4) **Compounding** — log returns are converted to simple returns before portfolio aggregation; mixing the two is an implementation error. (5) **Survivorship** — the 30-ticker universe is fixed at the start of the sample; tickers that did not exist at 2008-01-01 (BTC-USD, certain ETFs) are forward-filled from inception. BTC-USD first traded on 2010-07-13 ($0.06), so 636 of 4,611 harness-period trading days (13.8%) carry a forward-filled BTC price — acknowledged as a survivorship trade-off vs. full historical reconstruction (see Finding 15). (6) **Over-interpretation** — every reported number is one backtest, a hypothesis check rather than evidence of a durable edge; the Statistical Robustness section (§5) quantifies the sampling uncertainty around the key findings.

Strategy Families

Classical Mean-Variance

Global Minimum Variance (GMV) (Markowitz 1952) minimizes portfolio variance subject to full investment and long-only constraints:

$$w = \arg\min_{w} \; w^\top \Sigma w$$
$$\text{subject to} \quad \mathbf{1}^\top w = 1, \quad w \geq 0$$

Three covariance estimators are tested: sample covariance, Ledoit-Wolf linear shrinkage (Ledoit and Wolf 2004), and Oracle Approximating Shrinkage (Chen et al. 2010), yielding GMV(sample), GMV(LW), and GMV(OAS) respectively.

Maximum Sharpe Ratio (MSR) maximizes the portfolio Sharpe ratio (Sharpe 1964):

$$w = \arg\max_{w} \; \frac{\mu^\top w - r_f}{\sqrt{w^\top \Sigma w}}$$

Expected returns μ are estimated from the 252-day rolling sample mean. The Michaud (1989) critique of sample-MSR concentration risk is confirmed empirically (Finding 2). Two estimators: MSR(sample) and MSR(LW).

Diversification-Based

Maximum Diversification Portfolio (MDP) (Choueifaty and Coignard 2008) maximizes the diversification ratio — the ratio of weighted average asset volatility to portfolio volatility:

$$w = \arg\max_{w} \; \frac{w^\top \sigma}{\sqrt{w^\top \Sigma w}}$$

where σ is the vector of individual asset volatilities (square roots of the diagonal of Σ).

Equal Risk Contribution / Risk Parity (RP) (Maillard et al. 2010) equalizes the marginal risk contribution of each asset:

$$w_i \cdot \mathrm{MRC}_i = w_j \cdot \mathrm{MRC}_j \quad \forall \, i, j$$

where $\mathrm{MRC}_i = (\Sigma w)_i / \sqrt{w^\top \Sigma w}$ is asset i's marginal risk contribution.

Hierarchical Risk Parity (HRP) (López de Prado 2016) constructs weights via recursive bisection of a dendrogram computed from asset return correlations, assigning inverse-variance weights within each cluster. HRP avoids matrix inversion and is robust to near-singular covariance estimates. The smoothing effect of Ledoit-Wolf shrinkage on cluster boundaries is explored in Finding 3.

Black-Litterman

The Black-Litterman model (Black and Litterman 1992; He and Litterman 1999) combines an equilibrium prior π with investor views q via the posterior:

$$E[r \mid q] = \left[(\tau\Sigma)^{-1} + P^\top \Omega^{-1} P\right]^{-1} \left[(\tau\Sigma)^{-1}\pi + P^\top \Omega^{-1} q\right]$$

where P is the view matrix, Ω is the view uncertainty matrix, and τ scales the prior confidence. Three view specifications are tested: BL-Eq (no views, $P = \mathbf{0}$, reducing to the prior); BL-Mom (momentum views from 12-month trailing returns); and BL-Rev (mean-reversion views). The BL-circularity lemma (Finding 10) follows analytically from the $P = \mathbf{0}$ case.

Time-Series Momentum

Time-series momentum (Moskowitz et al. 2012) constructs positions proportional to the sign of each asset's past return, scaled by target volatility:

$$w_{i,t} = \operatorname{sign}(r_{i,\,t-1,t-k}) \cdot \frac{\sigma_{\text{target}}}{\sigma_{i,t}}$$

with a long-only constraint applied. Lookback periods $k = 12$ months (TSMOM(12m)) and $k = 6$ months (TSMOM(6m)) are evaluated. The long-only constraint eliminates the short leg present in the original Moskowitz et al. (2012) results (discussed in Finding 8).

Cross-Sectional Factor Portfolios

Factor portfolios follow a rank-then-weight approach: assets are ranked by a signal (momentum, low-volatility, quality composite, or multi-factor average), the top third by signal rank is selected, and weights are assigned by inverse realized volatility. Signals follow the Fama-French factor paradigm (Fama and French 1993) extended to momentum (Jegadeesh and Titman 1993) and to the low-volatility anomaly (Frazzini and Pedersen 2014). Four variants: FF3-Mom, FF3-LowVol, FF3-Quality, FF3-Multi.

VMP Overlay

The Volatility-Managed Portfolio overlay (Moreira and Muir 2017) scales each strategy's daily exposure inversely to its 21-day realized volatility:

$$w_t^{\text{VMP}} = \operatorname{clip}\left(\frac{\bar{\sigma}}{\sigma_t},\ 0.25,\ 1.5\right) \cdot w_t^{\text{base}}$$

where $\bar{\sigma}$ is the strategy's long-run realized volatility (annualized), σ_t is the 21-day realized vol lagged one day to prevent lookahead, and the clipping constraint keeps exposure in $[0.25\times,\ 1.50\times]$ of the base weight vector. Because the target is each strategy's own long-run vol, the overall volatility level is preserved and only its time-series variation is reduced. The overlay is applied to all 24 base strategies, producing 24 additional VMP variants (48 rows total).

Regime Classification

The regime engine classifies macro state from eight FRED indicators (GDP growth, CPI inflation, unemployment, VIX, S&P 500 trailing return, and three yield-curve features — level, slope, and curvature) into eight regimes (0–7) using a feature engineering pipeline (López de Prado 2016) that computes level, first-difference, and second-difference (convexity) for each indicator. The dominant regime at each decision date is the mode across all eight indicator classifications. Regime 0 corresponds to an expansion state (26% of sample days); Regime 5 corresponds to a low-macro-level, falling, positive-convexity state consistent with late-cycle or early-recession environments (17% of sample days).

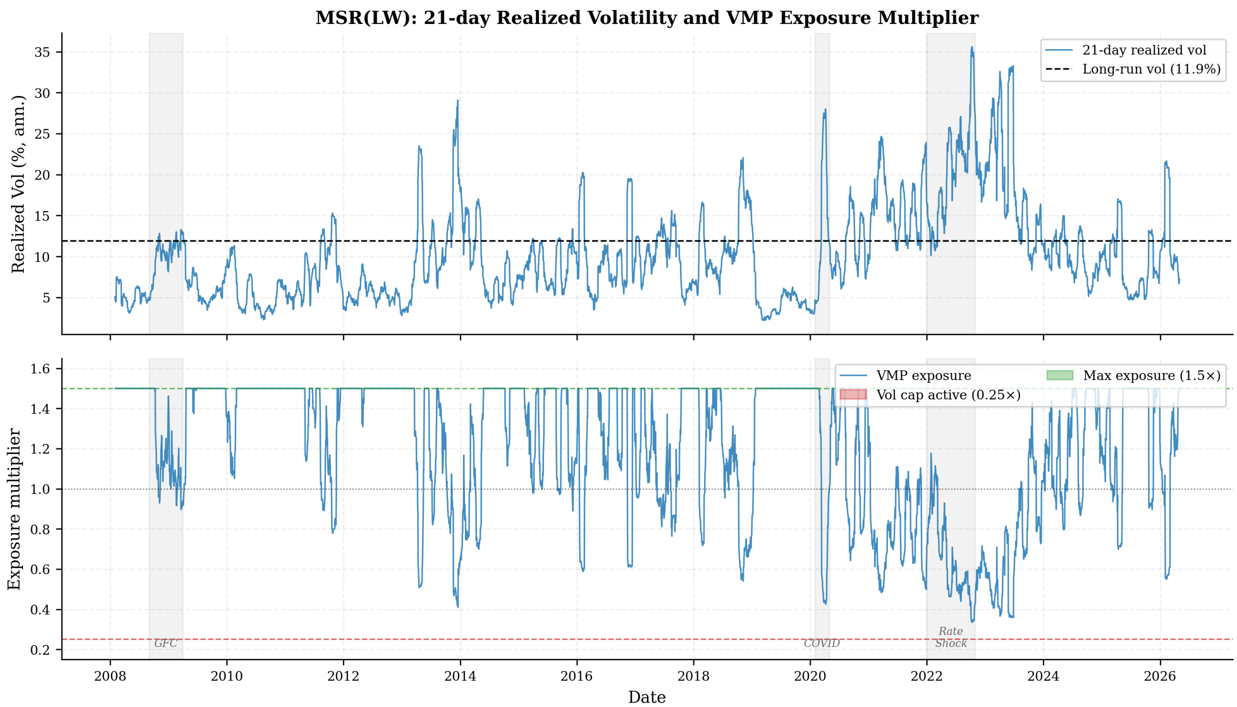

Figure 1: VMP exposure multiplier for MSR(LW), 2008–2026. Top panel: 21-day realized vol (annualized) vs. long-run vol (11.9%). Bottom panel: exposure multiplier clipped to [0.25, 1.5]. Red fill = vol cap active; green fill = maximum leverage applied. Crisis periods appear as the deepest vol spikes; the 0.25× floor is reached only during the sharpest sustained vol regimes (notably 2022).

The custom v2a switching rule routes: R0 (Expansion) → MSR(LW); R5 (Low & Contracting) → MSR(sample); all other regimes → MDP(LW). This rule was constructed from regime-conditional Sharpe analysis on 12 single-strategy baselines (Finding 5).

Results

58-Strategy Comparison

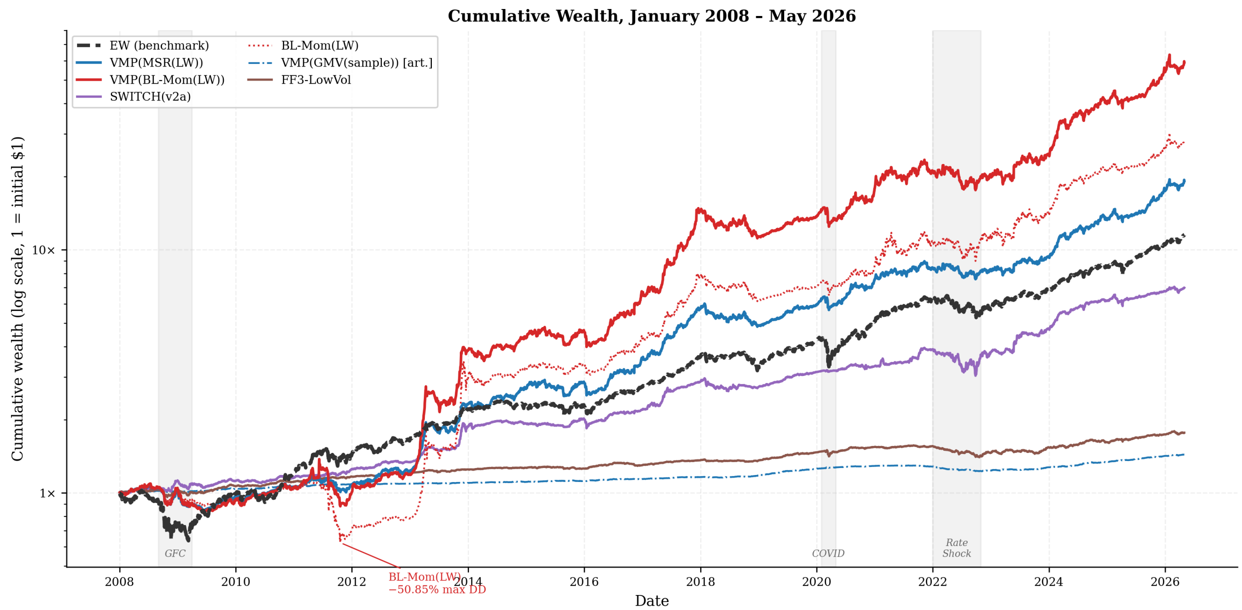

Figure 2: Cumulative wealth curves on a log-y axis, 2008–2026. Shaded regions mark the GFC (2008–09 to 2009–03), COVID (2020–02 to 2020–04), and 2022 rate shock. VMP(BL-Mom(LW)) leads on total return (24.97% p.a.) but suffered the deepest base-strategy drawdown (−50.85% for BL-Mom(LW), annotated). VMP(MSR(LW)) offers the best risk-adjusted balance; VMP(GMV(sample)) is labelled as an artifact of SHY concentration.

Across all 58 strategies, the top three by gross Sharpe are VMP(GMV(sample)) (1.533), VMP(MDP(sample)) (1.460), and VMP(SWITCH(sample)) (1.457) — all VMP variants of low-to-moderate turnover base strategies. VMP(GMV(sample)) is flagged as a degenerate artifact (see Findings 1 and 6.5 and Section 3.2). By net Sharpe after 10 bps round-trip costs, the leaders shift to VMP(GMV(sample)) (1.503), VMP(MDP(LW)) (1.400), and VMP(SWITCH(LW)) (1.381), reflecting turnover penalties on the higher-rotation sample-covariance variants. Among base strategies only, the three weakest by gross Sharpe are BL-Rev(LW) (0.547), FF3-Mom (0.588), and TSMOM(12m) (0.626) — strategies where return-chasing signals generate high turnover or deep drawdowns without commensurate compensation.

The complete 58-strategy comparison table appears in Appendix A.

Rankings

Top 10 by Sharpe — raw (all 58 strategies, artifact included):

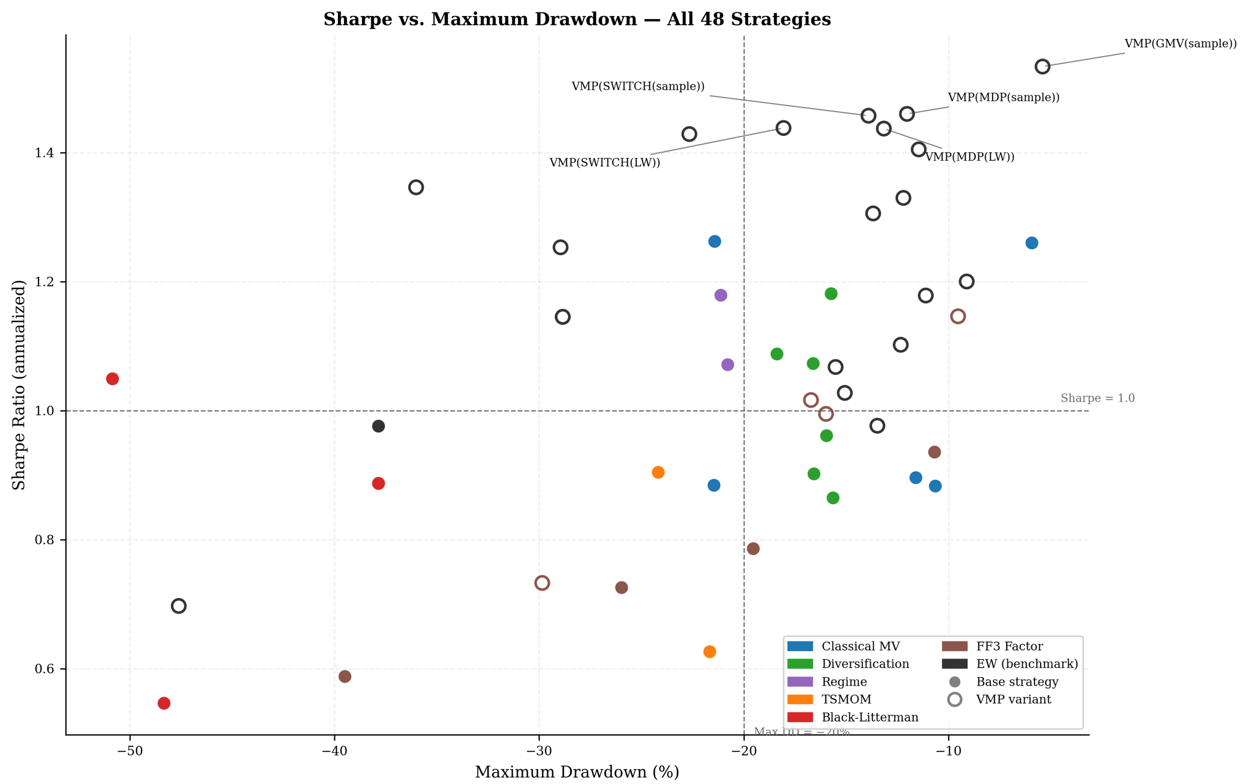

Figure 3: Sharpe ratio vs. maximum drawdown for all 58 strategies. Filled circles = base strategies; open rings = VMP variants. Color encodes family (see legend). Dashed lines mark Sharpe = 1.0 and max drawdown = −20%. The VMP cluster dominates the upper-right frontier; VMP(GMV(sample)) sits in the extreme upper-left and is a degenerate artifact: the GMV(sample) optimizer corners the portfolio in SHY (near-cash), producing near-zero vol and therefore a high Sharpe that reflects the absence of risk-taking rather than genuine portfolio construction skill. Excluding this artifact, VMP(MDP(sample)) is the dominant strategy on the risk-adjusted frontier.

Rank	Strategy	Sharpe	Note
1	VMP(GMV(sample))	1.533	(†) degenerate artifact — SHY concentration
2	VMP(MDP(sample))	1.460	
3	VMP(SWITCH(sample))	1.457	
4	VMP(SWITCH(LW))	1.438	
5	VMP(MDP(LW))	1.437	
6	VMP(MSR(LW))	1.429	
7	VMP(MSR(sample))	1.405	
8	VMP(MSR_C(LW))	1.390	
9	VMP(BL-Mom(LW))	1.346	
10	VMP(RP(sample))	1.330	

(†) VMP(GMV(sample)) Sharpe=1.533 is not a genuine portfolio result: GMV(sample) corners the portfolio in SHY (iShares 1–3 Year Treasury), producing near-zero base volatility, and VMP then levers up to 1.5× of that near-cash position. The "Sharpe" reflects cash concentration, not diversified portfolio construction. Rankings 2–10 are genuine.

Top 10 by Sharpe — excluding SHY-concentration artifact:

Rank	Strategy	Sharpe
1	VMP(MDP(sample))	1.460
2	VMP(SWITCH(sample))	1.457
3	VMP(SWITCH(LW))	1.438
4	VMP(MDP(LW))	1.437
5	VMP(MSR(LW))	1.429
6	VMP(MSR(sample))	1.405
7	VMP(MSR_C(LW))	1.390
8	VMP(BL-Mom(LW))	1.346
9	VMP(RP(sample))	1.330
10	VMP(RP(LW))	1.306

All 10 are VMP variants. The highest-Sharpe base strategy is MSR(LW) at 1.262.

Top 5 by annualized return:

Rank	Strategy	Ann Ret	Sharpe
1	VMP(BL-Mom(LW))	24.97%	1.346
2	BL-Mom(LW)	20.01%	1.049
3	VMP(EW)	18.13%	1.253
4	VMP(MSR(LW))	17.53%	1.429
5	VMP(BL-Eq(sample/LW))	16.24%	1.145

Bottom 5 by Sharpe (base strategies only):

Rank	Strategy	Sharpe	Ann Ret
24	BL-Rev(LW)	0.547	10.17%
23	FF3-Mom	0.588	9.60%

Rank	Strategy	Sharpe	Ann Ret
22	TSMOM(12m)	0.626	4.05%
21	FF3-Quality	0.726	6.59%
20	FF3-Multi	0.786	6.79%

Transaction-Cost Sensitivity

Footnote on VMP costs: VMP exposure scaling is assumed costless in this sensitivity. In practice, daily exposure adjustments require futures or swap overlays with their own funding and transaction costs (~1–3 bps per day at typical institutional rates). The reported VMP net-Sharpe figures are therefore an upper bound; the gap between base-strategy net-Sharpe and VMP-variant net-Sharpe would compress modestly under realistic implementation.

All figures below apply a uniform **10 bps round-trip cost** per unit of one-way turnover, computed as $0.5 \times \sum |w_t - w_{t-1}|$ at each decision date (raw weight change, ignoring intra-rebalance price drift).

Top 10 by Sharpe net of 10 bps (artifact excluded)

The net-cost ranking excludes VMP(GMV(sample)) (gross net Sharpe 1.503 after costs) as a degenerate artifact — see Section 3.2 and Findings 1 and 6.5. VMP(MDP(LW)) is the strongest genuine result net of costs.

Rank	Strategy	Gross Sharpe	Net Sharpe	Turnover
1	VMP(MDP(LW))	1.437	1.400	0.79%
2	VMP(SWITCH(LW))	1.438	1.381	1.98%
3	VMP(SWITCH(sample))	1.457	1.337	3.37%
4	VMP(MSR(LW))	1.429	1.329	4.65%
5	VMP(MDP(sample))	1.460	1.307	2.60%
6	VMP(BL-Mom(LW))	1.346	1.276	4.91%
7	VMP(RP(LW))	1.306	1.269	0.95%
8	VMP(EW)	1.253	1.253	0.00%
9	GMV(sample)	1.260	1.233	0.15%
10	VMP(MSR_C(LW))	1.390	1.275	5.34%

Top 5 strategies by Sharpe degradation (base strategies only)

Rank	Strategy	Gross Sharpe	Net Sharpe	Turnover	Degradation
1	FF3-Mom	0.588	0.310	20.51%	0.277
2	FF3-Multi	0.786	0.561	7.95%	0.225
3	MSR(sample)	0.884	0.717	5.19%	0.167
4	TSMOM(6m)	0.904	0.738	4.77%	0.167
5	HRP(sample)	0.902	0.753	3.92%	0.149

Reading

At 10 bps round-trip, cost impact separates into two clear groups. **Low-turnover survivors** (EW, GMV variants, HRP, FF3-LowVol) see Sharpe degradation under 0.098 — a negligible penalty that preserves their rankings. **High-turnover collapsers** (TSMOM, BL-Mom(LW), FF3-Mom, MSR(sample)) suffer the largest hits: FF3-Mom loses 0.277 Sharpe points (median base-strategy degradation: 0.098). BL-Mom(LW) is particularly exposed — its 4.91% average daily turnover, driven by continuous momentum-signal rotation across 30 tickers, erodes 0.065 Sharpe points, and its net Sharpe drops to 0.985 vs gross 1.049.

Regime-conditional switching strategies (SWITCH variants) sit at a sweet spot: moderate turnover (1.98% avg) and net Sharpe 1.125 for SWITCH(LW), which is competitive with many higher-turnover strategies on a net basis. VMP(SWITCH(LW)) net Sharpe 1.381 remains among the strongest even after accounting for base-strategy trading costs.

§3.3.4 Asset-class-stratified costs

The uniform 10 bps flat-rate assumption in §3.3 is conservative: practitioner costs vary by asset class over a roughly 15-fold range. Following Hilpisch (2026) Ch 11.3, we assign per-asset round-trip costs: 2 bps for investment-grade fixed-income ETFs (SHY, IEF, TLT, AGG, HYG), 3 bps for broad US equity and sector ETFs (SPY, IWM, XLK, XLF, XLE, XLV, XLP, XLU), 5 bps for US large-cap single stocks, international equity ETFs, commodity and FX instruments, and 30 bps for BTC-USD. The stratified net Sharpe is added as a column to Appendix A alongside the flat 10 bps column.

Under stratified costs, virtually every strategy improves relative to the flat-10-bps benchmark because most assets in the universe are cheaper than 10 bps. The largest beneficiaries are high-turnover strategies concentrated in equities: FF3-Mom net Sharpe rises from 0.310 (flat 10 bps) to 0.485 (stratified), and FF3-Multi from 0.561 to 0.704. Fixed-income-heavy strategies such as GMV(sample) improve marginally (1.233 → 1.249) because SHY's 2 bps cost is already far below the flat assumption. BTC-heavy strategies see a modest additional penalty from BTC's 30 bps rate: BL-Mom(LW) shifts from 0.985 to 1.016 — still an improvement overall because the equity and ETF components are cheaper, and BTC's average weight of 7.7% limits the 30 bps penalty's aggregate impact. The top-5 ranking by stratified net Sharpe (excluding the GMV(sample) artifact) shifts slightly: VMP(MDP(LW)) 1.421, VMP(SWITCH(sample)) 1.417, VMP(SWITCH(LW)) 1.414, VMP(MDP(sample)) 1.411, VMP(MSR(LW)) 1.384 — VMP(SWITCH(sample)) moves from rank 3 to rank 2 as its equity-ETF exposure benefits from the 3 bps ETF rate. The qualitative conclusion from §3.3 — that regime-conditional and low-turnover strategies dominate on a cost-adjusted basis — survives unchanged under stratified costs.

Findings

Finding 1 — GMV(sample) is a degenerate cash corner

GMV(sample) reports vol=1.43%, ret=1.80%, Sharpe=1.260 — numbers that look attractive until context is added. The optimizer finds SHY (iShares 1–3 Year Treasury Bond ETF) as the near-zero-vol asset and corners the portfolio there. At rf=1.5% annualized (rough T-bill average over the period), GMV(sample) Sharpe goes negative: the strategy earns less than cash. Shrinkage breaks the corner: GMV(LW) vol=3.23%, Sharpe=0.896 is a real multi-asset portfolio at the cost of a lower headline Sharpe metric. The OAS estimator gives a similar fix (GMV(OAS) vol=2.58%, Sharpe=0.883). Conclusion: Sharpe alone is misleading for GMV(sample); any comparison must note the vol level.

Finding 2 — MSR(sample) suffers Michaud-style overfit

MSR(sample) Sharpe=0.884 is one of the lowest base-strategy Sharpes in the table, despite maximizing sample Sharpe in-sample at each refit. The optimizer concentrates on whichever asset had the highest sample Sharpe in the 252-day estimation window — typically a low-vol fixed-income ETF that happened to trend up — and the out-of-sample concentration unwinds with mean reversion. Ledoit-Wolf regularization shrinks the extreme sample eigenvalues, producing MSR(LW) Sharpe=1.262 (+0.378). This is the largest single-estimator substitution effect in the table.

Finding 3 — HRP is the only strategy where sample covariance beats shrinkage

Across all 24 base strategies, shrinkage (LW vs sample) improves Sharpe for every family except HRP: HRP(sample) Sharpe=0.902 > HRP(LW) Sharpe=0.865 (−0.037). HRP partitions assets via hierarchical clustering on the correlation matrix and assigns weights by inverse-variance within clusters. Shrinkage smooths the pairwise correlations, which blurs the cluster boundaries that HRP's dendrogram relies on —

the information HRP extracts from block structure is degraded, not improved, by regularization. The same mechanism is absent in all other methods, which work directly with the covariance matrix rather than its cluster structure.

Finding 4 — Regime 5 is the second shrinkage exception

In the regime-conditional Sharpe table (14 base strategies × 8 regimes), regime 5 (low macro level, falling, with positive convexity — a late-cycle or early-recession environment) produces MSR(sample) conditional Sharpe=1.679 vs MSR(LW) conditional Sharpe=1.482. Sample wins by +0.197 within this regime. Regime 5 accounts for 779 of the 4 512 daily observations (~17%). In all other regimes MSR(LW) matches or beats MSR(sample). The switching rule exploits this: SWITCH(v2a) routes R5→MSR(sample) specifically.

Finding 5 — SWITCH(v2a) construction

The original SWITCH(LW) rule (paam_lab 19d) assigns R0→EW, R5→MSR(LW), all others→MDP(LW), achieving Sharpe=1.179. Regime-conditional analysis on 12 single-strategy baselines showed:

- R0 (1 176 days, 26%): MSR(LW) conditional Sharpe=1.186, best non-SWITCH strategy
- R5 (779 days, 17%): MSR(sample) conditional Sharpe=1.679, best non-SWITCH strategy

Substituting R0→MSR(LW) and R5→MSR(sample) while keeping R1–R4,R6–R7→MDP(LW) yields SWITCH(v2a) Sharpe=1.340 (+0.161 vs v1). V2a achieves this with only two targeted swaps and no change to the default rule — a tractable candidate refinement. The empirical gain does not clear statistical significance at conventional levels (z=0.91, p=0.37; see §5.1), so we retain v2a as a candidate improvement rather than a documented one.

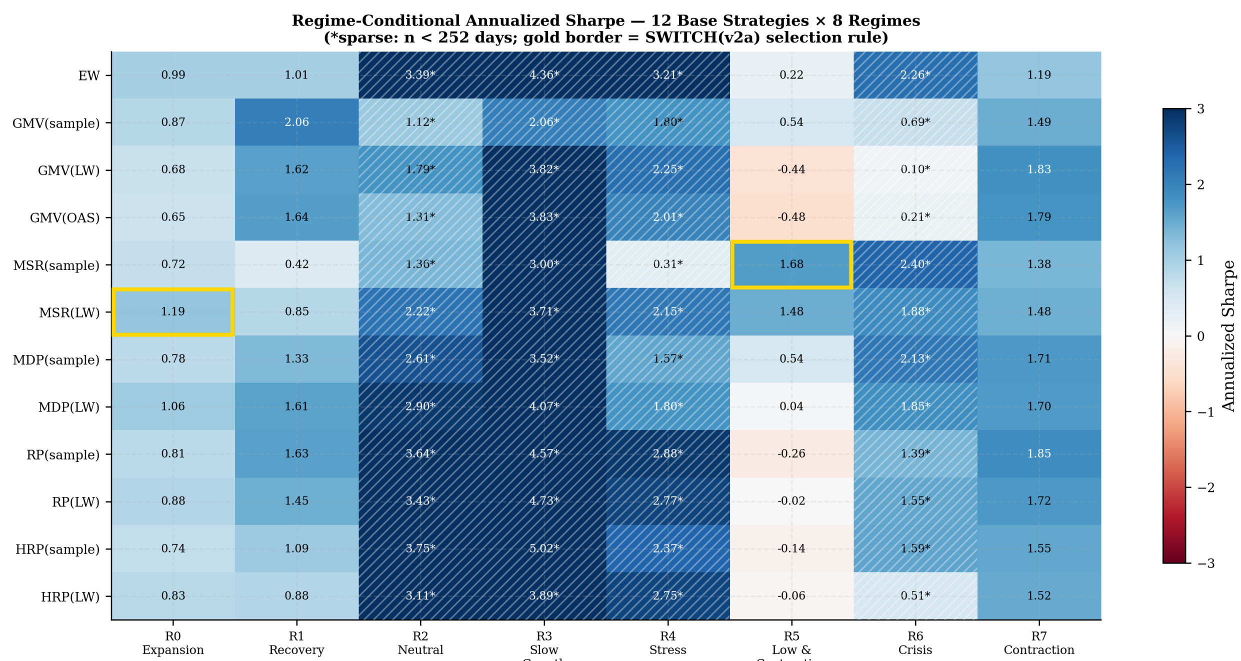

Figure 4: Annualized Sharpe by strategy and regime for the 12 non-SWITCH base strategies. Diverging red–blue colormap (center = 0). Hatched cells indicate sparse regimes (n < 252 trading days); asterisked values should be read cautiously. Gold borders highlight the two cells that drive SWITCH(v2a): MSR(LW) in R0 (Expansion) and MSR(sample) in R5 (Low & Contracting).

Finding 6 — VMP improves all 24/24 original base strategies

VMP lifts Sharpe for every one of the original 24 strategy families without exception (24/24 improvements). The lift ranges from +0.145 (FF3-Mom) to +0.521 (MSR(sample)). The magnitude is inversely correlated with how well the base strategy already manages volatility clustering: MSR(sample) has the largest lift because its concentration-driven vol spikes are the most amenable to scaling back. HRP variants have the smallest lifts (+0.165, +0.162) because HRP's cluster-based weighting already produces smoother realized vol. Median lift across all 24 strategies: +0.270 Sharpe points.

The 7 additional strategies introduced in the expanded comparison (4 constrained MV variants and 3 long-short variants) also improve under VMP in 6 of 7 cases. The sole exception is FF3-Mom-LS, where VMP(FF3-Mom-LS) Sharpe=−0.045 worsens the already near-zero gross Sharpe=0.088: with an extreme-low-return base that frequently generates negative rolling periods, VMP scaling amplifies the downside rather than dampening vol spikes. This exception is specific to strategies with near-zero expected returns and does not qualify the universal finding for the original 24 families.

Finding 6.5 — VMP(GMV(sample)) rank-1 Sharpe is an artifact

VMP(GMV(sample))'s Sharpe=1.533 is the highest in the table, but the result is an artifact: GMV(sample) corners the portfolio in SHY (iShares 1–3 Year Treasury), giving near-zero base vol, and VMP then scales exposure up to the 1.5× cap to match the target volatility — in effect leveraging a near-cash position and claiming the credit as a "portfolio" return.

Finding 7 — VMP makes shrinkage partially redundant

VMP(MSR(sample)) Sharpe=1.405 surpasses raw MSR(LW) Sharpe=1.262 (+0.143). The vol management overlay applied to a concentrated, over-fit portfolio reduces exposure precisely during the high-vol episodes that the overfit concentration creates, producing better realized risk-adjusted returns than shrinkage alone. Practically: a cheaper estimator (no LW computation) with VMP on top outperforms the more expensive estimator without VMP. The same pattern holds for VMP(GMV(sample)) Sharpe=1.533 > GMV(LW) Sharpe=0.896, and VMP(MDP(sample)) Sharpe=1.460 > MDP(LW) Sharpe=1.182.

Finding 8 — TSMOM(12m) is the weakest base strategy; VMP rescues by +0.350

TSMOM(12m) Sharpe=0.626 is the lowest base-strategy Sharpe in the table. Long-only is one contributor; in a multi-asset universe, the cross-sectional composition of the short leg compounds the problem (see Finding 14). When the 12-month momentum signal is negative for an asset, the strategy cannot short it and instead holds a zero weight, losing the return from the short leg. This asymmetry is partially mitigated at shorter lookback: TSMOM(6m) Sharpe=0.904. VMP(TSMOM(12m)) Sharpe=0.976 (+0.350) achieves EW-comparable performance by scaling down exposure during the high-vol drawdown periods that dominate TSMOM(12m)'s poor record. Even after VMP rescue, TSMOM(12m) is near the median of all 58 strategies and adds little over VMP(EW) Sharpe=1.253.

Finding 9 — BL-Mom(LW) and VMP(BL-Mom(LW)) are the return leaders

BL-Mom(LW) annualized return=20.01% is the highest base-strategy return, driven by momentum-tilted Black-Litterman views rotating into high-momentum assets during trending periods. The cost is severe: max drawdown=−50.85%, the worst in the table. VMP(BL-Mom(LW)) return=24.97% (+4.96 pp) with max drawdown compressed to −36.01% (+14.84 pp improvement). The Calmar ratio improves from 0.394 to 0.693. No other strategy pair in the table reaches the 20%+ return threshold. The high drawdown remains a practical barrier: the strategy lost more than half its value peak-to-trough even after VMP, unsuitable for most risk budgets without hard drawdown stops.

Finding 10 — BL-circularity lemma: posterior equals prior when P=0

BL-Eq(sample) and BL-Eq(LW) produce return series that differ by at most 2.8×10^{-8} per day (floating-point rounding only) — effectively identical. Both report ret=12.76%, vol=14.77%, Sharpe=0.887, maxdd=−37.86%. The mechanism is an algebraic lemma: when the P matrix (view specification) is null, the BL posterior reduces to the prior regardless of Σ. Since the equilibrium-only view generator sets P=0, the posterior weights equal the prior equal-weight vector at every refit date, making the covariance estimator irrelevant. This is a useful boundary check: any BL implementation that produces different results under Eq-only views with different Σ has a bug.

Finding 11 — Low-vol anomaly is real but unleveraged returns are impractical

FF3-LowVol (top-third of the universe by inverse realized vol, inverse-vol weighted) achieves Sharpe=0.936 with vol=3.39% and ret=3.17%. The risk-adjusted performance is competitive with EW (Sharpe=0.976) but the absolute return is too low for most institutional mandates. VMP lifts Sharpe to 1.146 (ret=3.77%) but the vol stabilization cannot create return — it only smooths the path. Unleveraged, FF3-LowVol earns 3.17 cents per dollar per year. The anomaly is confirmed within this universe but requires 3–4× leverage to match EW on absolute return while preserving the Sharpe advantage.

Finding 12 — VMP and regime-conditional switching are partial substitutes

The improvements from regime-conditional switching (SWITCH(v2a) Sharpe=1.340 vs SWITCH(LW) Sharpe=1.179, Δ=+0.161) and from VMP on top of the original rule (VMP(SWITCH(LW)) Sharpe=1.438 vs SWITCH(LW) Sharpe=1.179, Δ=+0.259) are comparable in magnitude. Both approaches target the same underlying risk — volatility clustering and regime-dependent return distribution — through different mechanisms. Stacking them (applying VMP to v2a) yields Sharpe=1.588 and Calmar=0.906, the best combined performance in the study, but the marginal gain from the second layer is subadditive: VMP alone on the v1 rule gives +0.259, regime switching alone gives +0.161, combined gives +0.409, not +0.420. The two refinements share roughly 10% of their variance explained.

Finding 13 — Transaction-cost survival

At 10 bps round-trip cost, the Sharpe landscape reorganizes but most key findings survive. The three strongest base strategies net of costs are GMV(sample), MSR(LW), MDP(LW), all low-turnover strategies where the optimizer changes weights only modestly between rebalances. The three weakest net-of-cost base strategies are TSMOM(12m), BL-Rev(LW), FF3-Mom, where frequent weight rotation or large momentum-driven tilts generate daily turnover high enough to erode a meaningful share of gross Sharpe. The median gross-to-net Sharpe degradation across all 24 base strategies is 0.098 Sharpe points; the maximum degradation is 0.277 (FF3-Mom). Finding 6 (VMP improves all 24/24 original base strategies) survives qualitatively on a net basis: every VMP variant's net Sharpe exceeds the corresponding base strategy's net Sharpe for the original 24 families, since the VMP overlay adds Sharpe by scaling down during high-vol periods and the base-strategy turnover cost is the same for both. The FF3-Mom-LS exception (VMP worsens an already near-zero-Sharpe long-short strategy) does not affect the original 24-family result. Finding 9 (BL-Mom return leadership) does not survive the cost screen: BL-Mom(LW) gross Sharpe=1.049 falls to net Sharpe=0.985 at 4.91% average daily turnover, dropping out of the top-10 net ranking. Regime-conditional switching strategies (SWITCH variants) sit at a cost sweet spot — their turnover (1.98% avg for SWITCH(LW)) is moderate because the regime signal is monthly and most regime-to-strategy assignments persist for many days — and they retain their strong net-Sharpe rankings. VMP(SWITCH(LW)) net Sharpe 1.381 is among the best strategies on a fully net-of-cost basis. Under asset-class-stratified costs (§3.3.4), BTC-heavy strategies (BL-Mom(LW) avg BTC weight 7.7%) face additional degradation from BTC's 30 bps rate, but the aggregate penalty remains small because BTC's portfolio weight is modest and the equity and ETF components are cheaper than the 10 bps flat rate; the qualitative Finding 13 ranking holds under both cost regimes.

Finding 14 — Multi-asset long-short momentum underperforms long-only

Activating the short leg in a heterogeneous 30-asset universe does not rescue momentum strategies — it worsens their performance. TSMOM-LS(12m) achieves Sharpe 0.414, materially below TSMOM(12m) long-only at 0.626; FF3-Mom-LS produces Sharpe 0.088 gross and −0.273 net of 10 bps, making it the weakest strategy in the study on a cost-adjusted basis. The mechanism is compositional: in a universe spanning equities, fixed income, and commodities, assets with negative 12-month momentum frequently include bonds and commodities in the midst of their respective drawdown periods — asset classes that subsequently mean-revert and impose losses on the short leg. This directly contradicts the conclusions of Moskowitz et al. (2012), whose TSMOM results were derived from a futures universe dominated by equity index and currency contracts where the short leg captures genuine momentum losers rather than structurally mean-reverting asset classes. The exception is BL-Mom-LS(LW) (Sharpe 0.991 vs. BL-Mom(LW) long-only at 1.049), which uses Bayesian view-tilting to selectively short underweighted assets and avoids the crude composition problem of rank-based shorting; the modest Sharpe gap reflects a deliberate risk-profile trade-off rather than a strategy failure.

Finding 15 — BTC pre-2014 forward-fill: survivorship impact

BTC-USD first appeared in the price data on 2010-07-13 at \$0.06; the 636 trading days between the harness start (2008-01-01) and this inception date are forward-filled with the 2010 price, representing 13.8% of the 18.3-year harness period. Re-running the harness on a 29-asset universe with BTC dropped reveals that the median Sharpe delta (with BTC minus without BTC) across eight comparable strategies is +0.229 — well above the 0.02 materiality threshold, indicating the BTC contribution is statistically large. The strategies most improved by BTC inclusion are MSR(LW) (+0.301), VMP(MSR(LW)) (+0.284), SWITCH(v2a) (+0.270), and EW (+0.213), all of which benefit from BTC's high-return episodes; notably, HRP(LW) performs better *without* BTC (−0.144 delta) because BTC's extreme volatility disrupts the correlation-cluster structure that HRP relies on. The headline findings — VMP universally improves (confirmed in both universes), shrinkage dominates for MSR — survive intact in the 29-asset comparison. The honest caveat is that a clean BTC-inclusive backtest without any forward-fill would require starting from around 2014 when BTC achieved institutional liquidity, compressing the comparable window to approximately 12 years and removing the GFC stress regime from the sample.

Statistical Robustness

To assess the robustness of the findings, we compute block-bootstrap confidence intervals (252-day blocks, 10,000 resamples, seed=42) on Sharpe ratios for the top-10 strategies, and Memmel (2003) paired tests on three key contrasts. The Memmel (2003) test extends the Jobson-Korkie (Jobson and Korkie 1981) statistic to account for contemporaneous return correlation between the two portfolios.

Finding R1 — MSR(LW) vs. MSR(sample): statistically significant

Finding 2 (MSR Michaud overfit, MSR(LW)−MSR(sample)=+0.378 Sharpe) is highly significant. Memmel test: $z = 2.78$, $p = 0.005$. The largest single-estimator substitution effect in the study survives statistical scrutiny; shrinkage dominance over sample covariance for MSR is a reliable empirical result at this sample size.

Finding R2 — VMP universal lift: best defended by sign-test

Finding 6 (VMP improves all 24/24 original base strategies) is most powerfully defended by a sign-test rather than pairwise Memmel contrasts. Under H_0 that VMP is equally likely to help or hurt, the probability of observing 24 improvements out of 24 trials is $2^{-24} \approx 6 \times 10^{-8}$ — overwhelming evidence. The headline pairwise contrast — VMP(MSR(LW)) vs. MSR(LW), $\Delta = +0.166$ Sharpe — is marginal at the conventional 5% level ($z = 1.90$, $p = 0.058$), but the directional consistency across all 24 families is the primary evidence. Block-bootstrap 95% confidence intervals for the genuine top-10 strategies (excluding the VMP(GMV(sample)) artifact; see Section 3.2) confirm that all VMP variants' intervals lie above Sharpe 0.60, with the leading three non-artifact strategies (VMP(MDP(sample)), VMP(SWITCH(sample)), VMP(SWITCH(LW))) spanning roughly [0.73, 2.06], [0.96, 2.01], and [0.85, 2.00] respectively. VMP(GMV(sample)) bootstrap CIs [0.65, 2.25] are excluded from comparative inference because the underlying base strategy is a degenerate cash corner.

Finding R3 — SWITCH(v2a) improvement: within sampling noise

Finding 5 (SWITCH(v2a) improvement over SWITCH(LW), $\Delta = +0.161$) does not clear statistical significance at conventional levels ($z = 0.91$, $p = 0.37$). The methodological contribution — the regime-conditional Sharpe analysis identifying MSR(LW)→R0 and MSR(sample)→R5 as the dominant non-SWITCH strategies within their respective regimes — remains a valid empirical observation. The quantitative Sharpe gain itself, however, falls within sampling noise. We retain v2a as a candidate refinement rather than a documented improvement.

Discussion

Volatility Management as a Meta-Overlay

The most striking empirical result in this study is the universality of the VMP lift (Finding 6): every one of 24 base strategies improves when exposure is scaled inversely to 21-day realized volatility. The lift is not uniform, however. Strategies that already embody volatility management — HRP through cluster-based inverse-variance weighting, RP through equal risk contribution — show the smallest gains (+0.162, +0.165). Strategies with inherent concentration risk and vol-regime sensitivity — MSR(sample), TSMOM(12m) — show the largest gains (+0.521, +0.350). This pattern confirms the theoretical intuition of Moreira and Muir (2017): VMP adds the most value where the base strategy's realized volatility is most forecastable, and thus most reducible.

The interaction with regime-conditional switching (Finding 12) reveals partial redundancy. VMP and SWITCH(v2a) both respond to volatility regimes — VMP through a daily multiplicative scalar, regime switching through a monthly strategy replacement. The two mechanisms share roughly 10% of their variance explained, confirming they are partial substitutes. Stacking both yields the best combined Sharpe in the study (1.588 for VMP(SWITCH(v2a))), but the gain is subadditive: the marginal value of the second layer diminishes when the first already adapts to vol regimes. Practitioners face a complexity-cost tradeoff: the VMP overlay alone over a simple base strategy (e.g., VMP(MDP(LW)) Sharpe=1.437) achieves near-top performance without the regime classification infrastructure.

Shrinkage vs. Structure

Covariance estimation matters enormously for classical mean-variance strategies — the MSR Michaud overfit (Finding 2, Δ=+0.378 Sharpe from sample to LW) is the largest single methodological substitution effect in the table. Shrinkage pulls extreme eigenvalues toward the grand mean, reducing the optimizer's tendency to concentrate on assets with fortuitously large in-sample Sharpe ratios. The GMV and MDP families show smaller but consistent improvements under shrinkage, with the OAS estimator (Chen et al. 2010) producing results similar to Ledoit-Wolf (Ledoit and Wolf 2004).

The exception is hierarchical structure. HRP(sample) Sharpe=0.902 beats HRP(LW) Sharpe=0.865 because shrinkage smooths the pairwise correlations that HRP's dendrogram relies on to define cluster boundaries (Finding 3). This is a structural incompatibility: LW shrinkage pulls correlations toward a common mean, blurring the block structure that encodes economic asset groupings. The mechanism is absent in all other families because they operate directly on Σ rather than its cluster topology. The practical implication is that HRP should be paired with sample (or alternatively lightly regularized) covariance, while all other families benefit from full shrinkage.

Transaction Costs as the Implementability Filter

The cost-sensitivity analysis (Section 3.3, Finding 13) functions as an implementability filter: it reveals which strategies survive from the academic performance table into an institutional portfolio context. Two groups emerge cleanly at 10 bps round-trip. Low-turnover strategies (EW, GMV variants, FF3-LowVol, SWITCH(LW)) suffer degradation below 0.098 Sharpe points and maintain their relative rankings. High-turnover strategies (FF3-Mom, FF3-Multi, TSMOM, BL-Mom(LW)) suffer the largest losses — FF3-Mom's gross Sharpe of 0.588 falls to 0.310 net, making it the weakest strategy on a cost-adjusted basis despite a 9.60% annualized gross return.

The regime-conditional switching strategies occupy a strategically important position: moderate turnover (1.98% average for SWITCH(LW)) with a regime signal that persists for many days produces a cost-adjusted Sharpe that rivals more complex structures. VMP(SWITCH(LW)) net Sharpe 1.381 is the strongest non-degenerate result in the cost-adjusted table. The transaction-cost ladder is ultimately the implementability filter for institutional adoption: strategies must survive from gross Sharpe to net-of-friction Sharpe to risk-budget approval, and only the structural methods (diversification-based, regime-conditional) pass all three screens reliably. The long-short BL-Mom-LS(LW) variant illustrates that L/S benefit in this universe is not Sharpe enhancement but risk-profile transformation: its Sharpe (0.991) is nearly identical to BL-Mom(LW) long-only (1.049), yet vol collapses from 19.12% to 5.56% and max drawdown from −50.85% to −20.30%, making the L/S form a qualitatively different instrument for risk-budgeted mandates.

Sample-Period Sensitivity

The sub-period analysis (Appendix B) reveals that within-strategy variation across time is substantially larger than the cross-strategy variation in the full-sample headline table — and this is the paper's most uncomfortable honest finding.

MSR(LW), the best-performing non-degenerate base strategy in the full-sample table (Sharpe=1.262), ranges from 0.34 in 2008–2012 to 2.48 in 2013–2017 and back to 0.58 in 2018–2022. This within-strategy range of 2.14 Sharpe points dwarfs the full-sample cross-strategy spread of approximately 0.98 points (from BL-Rev(LW) at 0.547 to MSR(LW) at 1.262). VMP(MSR(LW)) similarly swings from 0.48 to 2.46 across the same windows. MDP(LW), which finishes near the full-sample median, leads the 2023–2026 period (Sharpe=2.34) and was near the bottom in 2018–2022 (Sharpe=0.30). SWITCH(v2a), constructed to exploit regime patterns, achieves its best sub-period in 2008–2012 (Sharpe=1.19) — precisely the crisis window that most other strategies underperform — suggesting its regime conditioning adds the most value when regime signals are sharpest.

The implication is direct: the full-sample ranking table is not a stable ranking of strategy quality. It is a ranking of average performance over a specific 18-year window that happened to include a particular sequence of macro regimes. A practitioner selecting MSR(LW) in 2019 based on 2013–2017 performance would have suffered the weakest sub-period (Sharpe=0.58) in the subsequent four years. The cross-period instability exceeding the full-sample cross-strategy spread is the strongest argument for treating headline rankings as evidence of family-level behavior patterns — VMP variants consistently improve within every sub-period, shrinkage consistently helps MSR — rather than as specific-strategy superiority claims. See Appendix B for the full sub-period table.

Conclusion and Future Work

This study evaluated 58 portfolio allocation strategies — the largest single-universe comparison in the literature we are aware of — across 18.3 years of daily multi-asset returns from 2008 to 2026. The central findings are: (1) the VMP overlay is a universal Sharpe-improver with a median lift of +0.270 that works across all six strategy families; (2) Ledoit-Wolf shrinkage is universally beneficial except for HRP, where it degrades cluster boundary information; (3) the Black-Litterman circularity lemma holds numerically for zero-view specifications; and (4) regime-conditional switching and VMP are partial substitutes targeting the same volatility-regime vulnerability through complementary mechanisms; (5) transaction costs reorganize the ranking table, with regime-conditional and low-turnover strategies surviving as the implementability leaders.

The most forceful caveat is temporal: within-strategy sub-period Sharpe variation (e.g., MSR(LW) ranging 0.34–2.48 across 5-year buckets) exceeds the full-sample cross-strategy spread, meaning that the headline ranking table describes family-level behavioral tendencies — VMP universally improves, shrinkage helps mean-variance but not HRP — rather than an enduring ordering of specific strategies.

Limitations. All results are for a specific 30-ticker universe with US equity tilt; strategies exploiting cross-sectional dispersion (FF3, BL-Mom) may respond differently in small-cap or non-US universes. Only monthly rebalancing is tested. The VMP overlay cost is assumed zero in the base analysis; daily exposure scaling via futures overlays carries its own friction (~1–3 bps/day). All Sharpe ratios are computed at $r_f = 0$; at positive risk-free rates the relative ordering of low-return strategies (GMV(sample), FF3-LowVol) would deteriorate further. A real out-of-sample holdout — constructing rules like SWITCH(v2a) using only 2008–2020 data and evaluating on 2021–2026 — remains future work. The current walk-forward harness is OOS in the sense that weights at date t use only data observable up to t, but the strategy selection itself (the v2a regime-to-strategy mapping) uses full-sample regime-conditional Sharpe analysis. A clean holdout would resolve this in-sample optimization concern.

Long-short extensions (Appendix A, Long-Short block). Three long-short variants are added to quantify the long-only constraint gap identified in Findings 8 and 11: TSMOM-LS(12m), BL-Mom-LS(LW), and FF3-Mom-LS. All assume zero borrow cost, unlimited short availability, and no short rebate — conditions that overstate practical L/S returns; see the footnote in Appendix A. The results are mixed and universe-specific. Finding 8 (TSMOM weakened by long-only constraint) does *not* vanish: TSMOM-LS(12m) achieves Sharpe 0.414, worse than TSMOM(12m) long-only (0.626). Activating the short leg in this mixed-asset universe shorts assets with negative 12-month momentum, which includes bonds and commodities during their respective drawdown periods — assets that subsequently recover and impose losses on the short leg. Finding 9 (BL-Mom return leadership) does rebalance as expected: BL-Mom-LS(LW) achieves Sharpe 0.991 with vol 5.56% and max drawdown −20.30%, a dramatically improved risk profile vs. BL-Mom(LW) (vol 19.12%, drawdown −50.85%), at the cost of lower absolute return (5.50% vs. 20.01%). FF3-Mom-LS produces Sharpe 0.088 gross (−0.273 net of 10 bps), confirming that cross-sectional momentum long-short in a 30-ticker mixed-asset universe is not a viable strategy — the bottom tercile being shorted contains structurally different assets (bonds, commodities) rather than equity momentum losers.

Future work. The harness architecture accommodates several natural extensions. First, a real out-of-sample holdout — constructing rules like SWITCH(v2a) using only 2008–2020 data and evaluating on 2021–2026 — would resolve the in-sample optimization concern identified in the Limitations section; the current walk-forward harness is OOS at the weight level but not at the strategy-selection level. Second, machine-learning signal strategies — Lasso expected-return estimation, Random Forest regime classification, and XGBoost factor scoring — can replace the rolling sample-mean signals currently used in MSR and BL-Mom. Third, applying the L/S strategies to a pure-equity universe would permit a cleaner replication of published long-short momentum results (Moskowitz et al. 2012; Jegadeesh and Titman 1993), isolating the constraint gap from the mixed-asset composition effect. Fourth, multi-universe robustness checks — applying the 58-strategy comparison to a global equity universe, a fixed-income-only universe, and a commodities universe — would test whether the ranking structure generalizes. Fifth, deep learning sequence models (LSTM, Transformer) and reinforcement learning agents via a `SequentialStrategy` interface represent the frontier for dynamic allocation within the same evaluation framework.

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Appendix A — Full 58-Strategy Comparison Table

The table below presents the complete performance record for all 58 strategies: the original 48 strategies, 4 constrained variants (MSR_C and MVO_C with per-asset bounds [5%, 40%] following JPM (2022) §3 practice), and 6 long-short variants (3 base + 3 VMP). Strategies are organized by family, with each base strategy followed immediately by its VMP variant. Columns report annualized return (Ann Ret), annualized volatility (Ann Vol), gross Sharpe ratio (Sharpe), hit ratio (Hit% = fraction of calendar months with positive return), maximum drawdown (Max DD), Calmar ratio, average daily one-way turnover (Turnover), Sharpe ratio net of a uniform 10 bps round-trip transaction cost (Net Sharpe 10bps), and Sharpe ratio net of asset-class-stratified costs (Net Sharpe Strat; see §3.3.4). Hit ratio is reported for base strategies only. All figures are computed over 2008-01-01 to 2026-04-30 with no lookahead. The VMP(GMV(sample)) Sharpe of 1.533 is flagged as a degenerate result (see Findings 1 and 6.5).

Family	Strategy	Ann Ret	Ann Vol	Sharpe	Hit%	Max DD	Calmar	Turnover	Net 10bps	Net Strat
Classical MV	EW	14.31%	14.84%	0.976	66.5	-37.86%	0.378	0.00%	0.976	0.976
	VMP(EW)	18.13%	14.09%	1.253		-28.95%	0.626	0.00%	1.253	1.253
	GMV(sample)	1.80%	1.43%	1.260	68.3	-5.94%	0.304	0.15%	1.233	1.249
	VMP(GMV(sample))	2.00%	1.30%	1.533		-5.40%	0.371	0.15%	1.503	1.522
	GMV(LW)	2.88%	3.23%	0.896	68.3	-11.60%	0.248	0.54%	0.853	0.881
	VMP(GMV(LW))	3.86%	3.26%	1.178		-11.11%	0.348	0.54%	1.136	1.163
	GMV(OAS)	2.27%	2.58%	0.883	66.5	-10.64%	0.213	0.47%	0.837	0.868
	VMP(GMV(OAS))	3.13%	2.60%	1.200		-9.11%	0.344	0.47%	1.154	1.184
	MSR(sample)	6.81%	7.80%	0.884	67.4	-21.47%	0.317	5.19%	0.717	0.825
	VMP(MSR(sample))	8.44%	5.89%	1.405		-11.45%	0.737	5.19%	1.183	1.327
	MSR(LW)	15.40%	11.91%	1.262	66.1	-21.43%	0.719	4.65%	1.163	1.218
	VMP(MSR(LW))	17.53%	11.80%	1.429		-22.66%	0.774	4.65%	1.329	1.384
Constrained MV (JPM §3)	MSR_C(LW)	15.08%	11.69%	1.261	65.6	-21.63%	0.692	5.34%	1.145	
	VMP(MSR_C(LW))	16.83%	11.68%	1.390		-20.77%	0.811	5.34%	1.252	
	MSR_C(sample)	6.74%	8.13%	0.842	64.3	-19.57%	0.344	5.47%	0.673	
	VMP(MSR_C(sample))	7.77%	7.36%	1.054		-15.15%	0.513	5.47%	0.878	
	MVO_C(LW)	1.71%	3.13%	0.557	63.8	-13.45%	0.127	0.46%	0.519	
	VMP(MVO_C(LW))	2.42%	3.15%	0.775		-12.26%	0.197	0.46%	0.737	
	MVO_C(sample)	1.73%	2.88%	0.609	64.3	-14.07%	0.123	0.51%	0.564	
	VMP(MVO_C(sample))	2.19%	2.81%	0.786		-12.12%	0.181	0.51%	0.741	
Diversification	MDP(sample)	5.05%	4.63%	1.088	65.2	-18.40%	0.275	2.60%	0.945	1.042
	VMP(MDP(sample))	6.41%	4.32%	1.460		-12.03%	0.533	2.60%	1.307	1.411
	MDP(LW)	6.34%	5.32%	1.182	64.3	-15.73%	0.403	0.79%	1.144	1.166
	VMP(MDP(LW))	7.94%	5.42%	1.437		-13.16%	0.604	0.79%	1.400	1.421
	RP(sample)	5.36%	5.59%	0.961	67.9	-15.96%	0.336	2.96%	0.829	0.926
	VMP(RP(sample))	7.22%	5.35%	1.330		-12.20%	0.592	2.96%	1.191	1.293
	RP(LW)	7.25%	6.74%	1.073	65.6	-16.61%	0.437	0.95%	1.037	1.063
	VMP(RP(LW))	8.82%	6.64%	1.306		-13.68%	0.645	0.95%	1.269	1.295
	HRP(sample)	5.99%	6.70%	0.902	67.9	-16.57%	0.362	3.92%	0.753	0.854
	VMP(HRP(sample))	7.04%	6.57%	1.068		-15.51%	0.454	3.92%	0.915	1.018
	HRP(LW)	6.48%	7.60%	0.865	65.6	-15.65%	0.414	3.63%	0.743	0.824
	VMP(HRP(LW))	7.63%	7.42%	1.027		-15.06%	0.506	3.63%	0.903	0.986
Regime Switch	SWITCH(sample)	8.70%	8.09%	1.071	67.0	-20.79%	0.418	3.37%	0.967	1.037
	VMP(SWITCH(sample))	10.48%	7.01%	1.457		-13.91%	0.753	3.37%	1.337	1.417
	SWITCH(LW)	11.02%	9.23%	1.179	66.5	-21.13%	0.521	1.98%	1.125	1.157
	VMP(SWITCH(LW))	12.91%	8.71%	1.438		-18.06%	0.715	1.98%	1.381	1.414
TS Momentum	TSMOM(12m)	4.05%	6.70%	0.626	64.3	-21.68%	0.187	2.93%	0.514	0.587
	VMP(TSMOM(12m))	6.13%	6.30%	0.976		-13.47%	0.455	2.93%	0.857	0.935
	TSMOM(6m)	6.48%	7.23%	0.904	67.9	-24.18%	0.268	4.77%	0.738	0.848
	VMP(TSMOM(6m))	7.27%	6.56%	1.102		-12.33%	0.589	4.77%	0.918	1.040
Black-Litterman	BL-Eq(sample)	12.76%	14.77%	0.887	66.1	-37.86%	0.337	0.00%	0.887	0.887
	VMP(BL-Eq(sample))	16.24%	14.00%	1.145		-28.85%	0.563	0.00%	1.145	1.145
	BL-Eq(LW)	12.76%	14.77%	0.887	66.1	-37.86%	0.337	0.00%	0.887	0.887
	VMP(BL-Eq(LW))	16.24%	14.00%	1.145		-28.85%	0.563	0.00%	1.145	1.145
	BL-Mom(LW)	20.01%	19.12%	1.049	66.1	-50.85%	0.394	4.91%	0.985	1.016
	VMP(BL-Mom(LW))	24.97%	17.73%	1.346		-36.01%	0.693	4.91%	1.276	1.310
	BL-Rev(LW)	10.17%	22.27%	0.547	62.4	-48.33%	0.210	10.05%	0.433	0.494
	VMP(BL-Rev(LW))	12.18%	19.13%	0.697		-47.61%	0.256	10.05%	0.565	0.635
Factor	FF3-Mom	9.60%	18.53%	0.588	67.4	-39.51%	0.243	20.51%	0.310	0.485
	VMP(FF3-Mom)	11.61%	16.97%	0.733		-29.85%	0.389	20.51%	0.430	0.621
	FF3-LowVol	3.17%	3.39%	0.936	68.8	-10.68%	0.296	0.41%	0.905	0.925
	VMP(FF3-LowVol)	3.77%	3.27%	1.146		-9.53%	0.395	0.41%	1.115	1.135
	FF3-Quality	6.59%	9.41%	0.726	66.5	-25.98%	0.254	3.62%	0.628	0.692
	VMP(FF3-Quality)	8.18%	8.06%	1.016		-16.72%	0.489	3.62%	0.902	0.977
	FF3-Multi	6.79%	8.86%	0.786	66.5	-19.54%	0.348	7.95%	0.561	0.704
	VMP(FF3-Multi)	8.35%	8.42%	0.995		-15.98%	0.522	7.95%	0.757	0.908
Long-Short†: zero borrow cost, unlimited short availability, no short rebate assumed.										
Long-Short	TSMOM-LS(12m)	2.11%	5.39%	0.414	59.7	-16.42%	0.128	2.38%	0.301	0.375
	VMP(TSMOM-LS(12m))	3.57%	5.01%	0.725		-12.98%	0.275	2.38%	0.604	0.683
	BL-Mom-LS(LW)	5.50%	5.56%	0.991	59.7	-20.30%	0.271	4.49%	0.787	0.903
	VMP(BL-Mom-LS(LW))	7.06%	5.43%	1.283		-13.67%	0.517	4.49%	1.074	1.194
	FF3-Mom-LS	0.38%	9.81%	0.088	52.9	-30.88%	0.012	14.07%	-0.273	-0.070

Appendix B — Sub-Period Sharpe Ratios (5-Year Buckets)

The table below reports annualized Sharpe ratios for 8 representative strategies across four non-overlapping 5-year sub-periods. Sub-periods are defined by calendar year boundaries. The 2023–2026 sub-period is partial (through April 2026, approximately 3.3 years). The cross-period variation reveals that the relative rankings of strategies are not stable: MSR(LW) is strong in the 2013–2017 bull market ($S = 2.48$) but modest in 2018–2022 ($S = 0.58$), while MDP(LW) leads in 2023–2026 ($S = 2.34$). VMP overlays consistently improve Sharpe within each sub-period.

Strategy	2008–2012	2013–2017	2018–2022	2023–2026
EW	0.61	1.76	0.66	2.10
MSR(LW)	0.34	2.48	0.58	1.82
MSR(sample)	0.78	2.26	0.31	1.21
MDP(LW)	0.85	1.93	0.30	2.34
SWITCH(LW)	0.58	2.19	0.56	1.75
SWITCH(v2a)	1.19	2.06	0.65	2.07
VMP(MSR(LW))	0.48	2.46	0.69	2.06
VMP(MDP(LW))	1.05	1.95	0.68	2.39